Modelling In Mechanics

Solutions

Overview of Mechanics

Solution 2

Forces

Weight acts towards Earth. A normal reaction acts perpendicular to a surface. Friction opposes sliding or its tendency. Tension acts along a taut string.

In equilibrium, resultant force is zero. For a particle of constant mass, Newton's second law connects force and motion:

$\mathbf{F} = m\mathbf{a}$.

Motion

Gradient of a displacement–time graph gives velocity. Gradient of a velocity–time graph gives acceleration. Signed area under velocity–time gives displacement; area under speed–time gives distance.

For constant acceleration:

$v = u + at, \quad s = ut + \frac{1}{2}at^2$

$v^2 = u^2 + 2as, \quad s = \frac{1}{2}(u + v)t.$

For variable acceleration:

$v = \dfrac{ds}{dt}, \qquad a = \dfrac{dv}{dt}.$

Modelling Assumptions

Solution 3

Common modelling assumptions

Particle: dimensions are negligible; mass is concentrated at a point. Rotation is ignored. Air resistance, when neglected, is a separate assumption.

Smooth surface: no friction. A rough surface can exert friction.

Light, smooth pulley: negligible mass and no friction; an ideal taut light string has equal tension on both sides.

Inextensible string: does not stretch. In a simple fixed-pulley system, connected particles have equal acceleration magnitudes while the string remains taut.

Rod: a rigid body whose transverse dimensions are negligible compared with its length.

Peg or support: fixed; dimensions may be neglected. Whether it is smooth or rough depends on the model.

Always state the assumption that justifies a simplification.

SI units

Solution 4

SI units

Quantity	Unit	Symbol
Mass	kilogram	kg
Displacement	metre	m
Time	second	s
Velocity	metres per second	$\mathrm{m\,s^{-1}}$
Acceleration	metres per second squared	$\mathrm{m\,s^{-2}}$
Force	newton	N

Newton's second law gives

$1\,\mathrm{N} = 1\,\mathrm{kg\,m\,s^{-2}}$.

Convert measurements into compatible units before substituting into an equation. Mass is measured in kilograms; weight is a force measured in newtons.

Scalars and vectors

A scalar has magnitude only. Distance and speed are non-negative; mass and time are scalar quantities.

A vector has magnitude and direction. Displacement, velocity, acceleration and force are vectors.

A displacement of $3\mathbf{i} + 4\mathbf{j}$ m has magnitude 5 m.

A particle moves 3 m opposite to the chosen positive direction. State its distance travelled and its displacement.

Distance and displacement

Distance $= 3$ m; displacement $= -3$ m.

The sign specifies direction. A one-dimensional vector can be written as a signed number.

1. A displacement has magnitude 5 m and is directed 60° above the positive horizontal axis. Find its components.

1

$\mathbf{s} = 5\cos 60^\circ \mathbf{i} + 5\sin 60^\circ \mathbf{j}$

$= (2.5\mathbf{i} + 4.33\mathbf{j})$ m (3 s.f.).

2. Velocity is $(5\mathbf{i} - 12\mathbf{j})\,\mathrm{m\,s^{-1}}$. Find the speed.

2

$|\mathbf{v}| = \sqrt{5^2 + (-12)^2} = 13\,\mathrm{m\,s^{-1}}$.

3. A force of 8 N acts 45° below the positive horizontal axis. Find its vector form.

3

$\mathbf{F} = (8\cos 45^\circ)\mathbf{i} - (8\sin 45^\circ)\mathbf{j}$

$= 4\sqrt{2}\mathbf{i} - 4\sqrt{2}\mathbf{j}$ N.

Further Examples

Solution 7

1. A force of 6 N acts downwards and to the left, at 60° to the downward vertical. Find its components.

1

$\mathbf{F} = -6\sin 60°\mathbf{i} - 6\cos 60°\mathbf{j}$

$= -3\sqrt{3}\mathbf{i} - 3\mathbf{j}$ N.

2. Find the magnitude of acceleration $(6\mathbf{i} - 8\mathbf{j})\,\mathrm{m\,s^{-2}}$.

2

$|\mathbf{a}| = \sqrt{6^2 + (-8)^2} = 10\,\mathrm{m\,s^{-2}}$.

3. A displacement of 4 m is directed 30° above the negative horizontal axis. Write it using unit vectors.

3

$\mathbf{s} = -4\cos 30°\mathbf{i} + 4\sin 30°\mathbf{j}$

$= -2\sqrt{3}\mathbf{i} + 2\mathbf{j}$ m.

Test Your Understanding

Solution 8

1. A man walks from A to B, then to C. His displacements are $\overrightarrow{AB} = (6\mathbf{i} + 4\mathbf{j})$ m and $\overrightarrow{BC} = (5\mathbf{i} - 12\mathbf{j})$ m.

   a. Find the magnitude of $\overrightarrow{AC}$.

   b. Find the total distance walked.

1a

$\overrightarrow{AC} = 11\mathbf{i} - 8\mathbf{j}$ m

$|\overrightarrow{AC}| = \sqrt{185} = 13.6$ m (3 s.f.).

1b

$d = \sqrt{6^2 + 4^2} + \sqrt{5^2 + 12^2}$

$= \sqrt{52} + 13 = 20.2$ m (3 s.f.).

2. A car's velocity is $(3\mathbf{i} - \mathbf{j})\,\mathrm{m\,s^{-1}}$. Find the angle its direction makes with $\mathbf{i}$.

2

$\theta = \tan^{-1}(1/3) = 18.4°$ (3 s.f.).

The direction is below the positive horizontal axis.

2. A car has velocity $(12\mathbf{i} - 10\mathbf{j})\,\mathrm{m\,s^{-1}}$. Find its speed and the angle with $\mathbf{i}$.

2

$|\mathbf{v}| = \sqrt{244} = 15.6\,\mathrm{m\,s^{-1}}$

$\theta = \tan^{-1}(10/12) = 39.8°$.

3. A motorbike has acceleration $(3\mathbf{i} - 4\mathbf{j})\,\mathrm{m\,s^{-2}}$. Find its magnitude and the angle with $\mathbf{j}$.

3

$|\mathbf{a}| = 5\,\mathrm{m\,s^{-2}}$

$\theta = 180° - \tan^{-1}(3/4) = 143°$ (3 s.f.).

7. A boat has acceleration $(-0.05\mathbf{i} + 0.15\mathbf{j})\,\mathrm{m\,s^{-2}}$. Find its magnitude and the angle with $\mathbf{i}$.

7

$|\mathbf{a}| = \sqrt{0.025} = 0.158\,\mathrm{m\,s^{-2}}$

$\theta = 180° - \tan^{-1}(3) = 108°$ (3 s.f.).